

MADELINE MARGARET TRELA BERGER

mmtb@hawaii.edu | [Github](#)

EDUCATION

- PhD Student Marine Biology *Expected Spring 2027*
Advisor: Dr. Mary Donovan, University of Hawai‘i Manoa (UHM)
- MS Conservation Planning Specialization & Environmental Data Science (2020)
Bren School of Environmental Science & Management, University of California, Santa Barbara (UCSB)
- BA Economics & Minor in Environmental Systems and Society (2016)
University of California, Los Angeles (UCLA)

RESEARCH AND PROFESSIONAL EXPERIENCE

- 2023 - **Graduate Research Assistant**
Hawai‘i Monitoring and Reporting Collaborative, UHM
(Supervisor: Dr. Mary Donovan)
- 2022 - 2023 **Senior Research Specialist**
Hawai‘i Monitoring and Reporting Collaborative, Arizona State University
(Supervisor: Dr. Mary Donovan)
- 2021 - 2022 **Marine Spatial Planning (MSP) Project Analyst**
McClintock Lab, National Center for Ecological Analysis and Synthesis
(Supervisor: Dr. Will McClintock)
- 2020 - 2021 **Spatial Analyst and Modeler**
National Center for Ecological Analysis and Synthesis (Supervisor: Dr. Ben Halpern)
- 2020 - 2020 **Ocean Health Index Fellow**
National Center for Ecological Analysis and Synthesis (Supervisor: Dr. Ben Halpern)
- 2019 - 2020 **Graduate Research Assistant (Master’s level)**
The Bren School of Environmental Science and Management, University of California Santa Barbara (Supervisor: Dr. Kelsey Jack)
- 2019 **Indigenous Peoples and Local Communities Mapping Intern,**
Conservation International (Supervisor: Dr. Sushma Shrestha)
- 2017 - 2018 **Food Systems Resilience Intern**
Natural Capital Project, Stanford University (Supervisor: Dr. Rebecca Chaplin-Kramer)

TEACHING & MENTORING

- 2025 **Co-Instructor:** “Software Carpentries: R for Ecologists”,
- 2024 **Co-Organizer:** R-Ladies Honolulu
- 2024 **Mentor** for undergraduate intern at Hawaii Pacific University

- 2023 **Certified Instructor**, The Carpentries
- 2021 **Co-Instructor** (with Dr. Will McClintock), “Marine Spatial Planning with SeaSketch”, (3-day workshops in Suva, Fiji and Apia, Samoa)
- 2020 **Co-Instructor** (with Dr. Will McClintock), “Introduction to Marine Spatial Planning”, (Online short course for Maldivian students), Scripps Institute of Oceanography
- 2019 **Graduate Teaching Assistant**: “Advanced Research Methods”, Department of Psychology, University of California Santa Barbara
- 2019 **Graduate Teaching Assistant**: “Statistics for Economics”, Department of Economics, University of California Santa Barbara
- 2018 **Graduate Teaching Assistant**: “History of Economics”, Department of Economics, University of California Santa Barbara

PUBLICATIONS

Berger, M., McClintock, W., Burt, C., Welch, T. (2023). SeaSketch. In: Burnett, C.M. (eds) Evaluating Participatory Mapping Software. Springer, Cham.

Berger, M., Canty, S. W., Tuholske, C., & Halpern, B. S. (2022). Sources and discharge of nitrogen pollution from agriculture and wastewater in the Mesoamerican Reef region. *Ocean & Coastal Management*, 227, 106269.

SCHOLARSHIPS AWARDED

- 2025 Colonel Willys E. Lord, DVM & Sandina L. Lord Scholarship Fund
Hawaii Institute of Marine Biology Student Scholarship
- 2025 The Wayne and Anita Baldwin Scholarship Fund
Hawaii Institute of Marine Biology
- 2025 Peter Castro Graduate Student Research Fund Scholarship Fund
Hawaii Institute of Marine Biology

PRESENTATIONS

- 2025 **American Fisheries Society Pacific Chapter Meeting**: “*Understanding concurrent patterns of land-based impacts and tropical nearshore fisheries in Hawai‘i*”, Honolulu, HI
- 2024 **American Fisheries Society Annual Meeting**: “*Mapping intersection of land-based stressors and nearshore fisheries for effective ecosystem management in Hawai‘i*”, Honolulu, HI
- 2024 **American Association of Geographers Annual Meeting**: “*Modeling and mapping human activities affecting nearshore conditions in the Main Hawaiian Islands*”, Honolulu, HI
- 2020 **Public Master’s Capstone Presentation**: “*Examining the impacts of circular supply chain through the lens of food waste*”, Santa Barbara, CA

CERTIFICATIONS, TRAINING, SKILLS

Languages: Non-native-speaker fluency in Spanish, native English speaker

Software: R statistical software, Python, JavaScript, GDAL/OGR, Bash & zsh shell, Google Earth Engine, GitHub, ArcGIS (Pro, ArcMap), QGIS, Mapbox, Microsoft Office, Adobe Creative Suite

Diving: AAUS Scientific Research Diver, CPR/Oxygen/ First Aid, Advanced Open Water Diver